COURSE PROGRAM

Control Valve Engineering Course

A practical, 32-hour course covering control valve fundamentals, sizing, selection and engineering project documentation.

32 HOURS	4 DAYS	8 MODULES

Who this course is for

Process specialists and engineers, instrumentation and control engineers, and mechanical engineers.

Before you start

Basic knowledge of process engineering and instrumentation.

Your instructor

Eng. R. Mahjoub
More than 24 years of experience across various industries, particularly in oil and gas projects.

The final project

Size, select, document and defend three valves, bringing together the calculations and project deliverables covered in the course.

Suggested four-day learning plan

01 Build the foundation

02 Turn process data into sizing

03 Select for the service

04 Bring it into the project

This document is a course outline, not a full lesson manual. The four-day allocation below is suggested; total course duration is 32 hours.

DAY 01 / 8 HOURS / SUGGESTED ALLOCATION

Build the foundation

MODULE 01

Control Valve Fundamentals

Topics covered

- Control valve function and operating principles
- Gate, Globe, rotary and Diaphragm valves
- Application of different valves.
- Single-port and double-port designs
- Valve body, trim, stem, packing, bonnet, and seat
- Normally open, normally closed, and fail-safe positions

MODULE 02

Flow Characteristics and Control Performance

Topics covered

- Inherent versus installed characteristics
- Linear, equal-percentage, and quick-opening trims
- Rangeability and turndown
- Valve authority and controllability
- Cavitation, flashing, noise, and vibration
- Effects of piping reducers and fittings

DAY 02 / 8 HOURS / SUGGESTED ALLOCATION

Turn process data into sizing

MODULE 03

Process Data and Sizing Fundamentals

Topics covered

- Required process inputs and operating cases
- Liquid, gas, steam, and two-phase flow considerations
- Density, specific gravity, viscosity, molecular weight and compressibility
- Flow rate, inlet pressure, outlet pressure, temperature and pressure drop
- Unit conversions and dimensional consistency
- Minimum, normal, maximum and start-up cases

MODULE 04

Control Valve Sizing Calculations

Topics covered

- Liquid sizing using Cv and Kv
- Gas and steam sizing
- Choked flow and critical pressure ratio
- Liquid pressure recovery and piping geometry factors
- Reynolds number and viscosity correction
- Required versus selected valve capacity
- Manual calculations versus software verification

DAY 03 / 8 HOURS / SUGGESTED ALLOCATION

Select for the service

MODULE 05

Valve, Actuator and Accessory Selection

Topics covered

- Valve body size versus line size
- Body and trim materials, pressure class and end connections
- Pneumatic diaphragm and piston actuators
- Spring return, air failure and position failure
- Smart positioners, I/P converters, solenoid valves and limit switches
- Actuator thrust, torque and seating requirements
- Control valve versus ESD valve requirements

MODULE 06

Advanced Applications and Standards

Topics covered

- Anti-cavitation and low-noise trim
- High-pressure-drop and severe-service applications
- Cryogenic, corrosive, erosive and high-temperature services
- Control valve noise prediction and mitigation
- IEC 60534 and ISA 75.01, ... guidance
- Pressure-temperature ratings and applicable material standards
- Inspection, testing and documentation requirements

DAY 04 / 8 HOURS / SUGGESTED ALLOCATION

Bring it into the project

MODULE 07

Sizing Software Workshop

Topics covered

- Create a valve sizing case
- Enter process conditions and fluid properties
- Run minimum, normal and maximum operating cases
- Review calculated capacity, opening, velocity and noise
- Select valve series, trim, actuator and accessories
- Generate and check sizing reports
- Investigate common sizing errors

MODULE 08

EPC Project Deliverables and Capstone

Topics covered

- Control valve datasheets and specifications
- Valve sizing calculation sheets
- Technical bid evaluation (TBE)
- Vendor document review
- P&ID; and instrument index coordination
- Final project: size, select, document and defend three valves